Shivam Balasaheb Bhagat

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PROFILE

MSc Advanced Computer Science student at the University of Manchester, with hands-on experience building backend systems, full-stack web applications, and AI/ML pipelines. MSc dissertation delivered a four-layer CI/CD documentation tool reaching 95.5% fact-recall against a 53.2% naive-LLM baseline across 201 pre-registered facts. Also built a full-stack novel-publishing platform (React 18, TypeScript, Supabase) and a RAG-based conversational system reaching 92.4% Recall@3. Comfortable across the stack: Python, JavaScript/TypeScript, REST APIs, PostgreSQL, Docker. Available for full-time software engineering roles in the UK from November 2026. Graduate Route right to work, no sponsorship required.

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript (ES6+), TypeScript
Back-end/DB: Node.js, REST APIs, PostgreSQL, MySQL, MongoDB, Supabase
Front-end: React, Next.js, HTML5, Tailwind CSS
Tools: Git/GitHub, Docker, VS Code, Linux, Vite, Vercel, GitHub Actions, Mermaid.js
Data/ML: PyTorch, scikit-learn, Pandas, NumPy, OpenCV, Transformers, LangGraph, FAISS, Streamlit, LLM APIs, Prompt Engineering

EDUCATION

University of Manchester — MSc Advanced Computer Science *Sep 2025 – Sep 2026*

- Global Talent Scholarship recipient; Academic Representative facilitating student–faculty communication.
- Dissertation submitted; all coursework complete — results expected November 2026, degree conferral December 2026.
- Relevant modules: Transforming Text into Meaning (NLP/LLMs/RAG), Security & Privacy in Artificial Intelligence, Software Security, Topics in Machine Learning.

Savitribai Phule Pune University — B.E. Computer Engineering *Sep 2021 – Jun 2025*

- CGPA: 8.35/10 — First Class. Final-year project: Ask the Doc (Generative-AI document assistant).

PROJECTS

"What is my CI Pipeline Doing?" — Automated CI/CD Documentation (MSc Dissertation)
Python · LLM APIs · GitHub Actions · YAML · Mermaid.js

- Built a four-layer tool generating automatic documentation for GitHub Actions pipelines: a parser converts workflow YAML into a platform-agnostic intermediate representation, then deterministic generators and a constrained LLM layer (restricted to extracted, structurally-validated facts) render readable Markdown/Mermaid output.
- Designed a four-tier evaluation framework benchmarked against a naive baseline (raw workflow YAML sent directly to an LLM with no parsing or extraction); pre-registered 201 objectively-checkable facts across 10 held-out pipelines of up to 40 jobs each.
- Achieved 95.5% fact-recall vs. 53.2% for the baseline, with error injection confirming the model layer altered wording, not facts; supervised by Professor Suzanne Embury, University of Manchester.

South Asian Culinary RAG Assistant (Group Project — 4 members)
LangGraph · FAISS · BGE-Large Embeddings · Qwen2.5 · Streamlit · Python

- Owned the retrieval core of a fully local RAG chatbot: built the FAISS vector store, BGE-Large embedding pipeline, and a LangGraph intent-routing state machine spanning seven intent categories.
- Delivered 92.4% Recall@3, 93% intent-classification accuracy, and 88.6% generation faithfulness across a 100-query evaluation benchmark.
- Combined semantic retrieval (1024-dim embeddings, L2 thresholding) with LLM-powered generation, exposed through a Streamlit dark-mode UI with live pipeline inspection (intent, entities, raw FAISS scores).

StoryWanderer — Web Novel Publishing Platform (in development)
React 18 · TypeScript · Vite · Tailwind CSS · shadcn/ui · Supabase

- Designed and built a full-stack novel-publishing platform end-to-end, from schema design to a Vercel deployment, handling authentication, chapter-level content management, and real-time data sync via Supabase/PostgreSQL.
- Secured the data layer with Postgres row-level security and a dedicated security view for comment authorship, and added stored tsvector columns for full-text search.
- Built a component-driven frontend in React 18, TypeScript, and shadcn/ui, including a custom long-form reading UI optimised for serialised fiction.

Fashion-MNIST CNN Classifier — Constrained Deep Learning
PyTorch · Docker · Python

- Engineered a compact 2-block CNN under a strict 100K-parameter budget (50,378 params), reaching 92.35% validation and 91.72% test accuracy on Fashion-MNIST.
- Ran a disciplined ablation study across four configurations — baseline CNN, CNN+Transformer hybrid (90.66%), data augmentation, and extended training — to select the best-generalising model rather than the highest raw accuracy.

- Trained and evaluated fully offline inside a network-isolated Docker container, using ReduceLROnPlateau scheduling and BatchNorm to reach stable convergence within 15 epochs.

Real-Time Driver Drowsiness Detection
YOLOv5 · PyTorch · OpenCV · NumPy

- Built a real-time fatigue-detection system using a custom-trained YOLOv5 model to classify awake/drowsy eye states from a live webcam feed at sustained frame rates, on a custom dataset labelled across variable lighting conditions.
- Implemented debounce logic to suppress false-positive alerts, balancing responsiveness against reliability for a safety-relevant use case.

Ask the Doc — Generative AI Document Assistant
Streamlit · HuggingFace Transformers · Pegasus · RoBERTa · gTTS · Python

- Built an end-to-end document-intelligence app combining abstractive summarisation (Pegasus) with extractive QA (RoBERTa-SQuAD2), plus text-to-speech output via gTTS.
- Final-year undergraduate project; architected with a modular pipeline separating ingestion, inference, and rendering layers for maintainability.

WORK EXPERIENCE

Online Order Picker (Dotcom) — Tesco, Altrincham *Sep 2026 – Present*

- Pick items for customer online orders, moving around the whole store to find and collect each product accurately.
- Check the warehouse when an item isn't on the shop-floor aisle, and escalate to the floor manager when it still can't be found.
- Rearrange and restock products on the aisles before rush hour; guide customers to products or find items for them.

Sandwich Artist / Team Member — Subway, Manchester *May 2026 – Oct 2026*

- Prepared the full menu to order during high-volume rush periods, working in a 3-person team to keep service fast and accurate.
- Operated the till (cash and card), and replenished and rotated stock to keep products fresh and well presented.
- Carried out store opening and closing procedures and kept the store to food safety and hygiene standards (Food Safety & Allergens certified).

LEADERSHIP & VOLUNTEERING

IEEE — Student Member & Event Organiser *Oct 2022 – Mar 2024*

- Co-organised technical workshops and AI/ML events; managed scheduling, venue coordination, and team logistics.

Volunteer — Saday Sadev NGO *Mar 2024 – Apr 2025*

- Produced social media content to increase programme awareness; drafted cross-organisation collaboration proposals.

National Service Scheme (NSS) — Member *Aug 2022 – Feb 2024*

- Organised a blood donation camp, supported community cleanliness drives, and volunteered at elderly care homes.

CERTIFICATIONS

ISC2 CC (Cybersecurity) 2024 · Python – NPTEL 2024 · Version Control – Coursera 2023 · Web Development – Udemy 2025